

Noonan Syndrome 6 (RAF1-Related): Comprehensive Disease Characterization

Disease: Noonan Syndrome 6 (NS6) **MONDO ID:** MONDO:0013186 | **OMIM:** #613224 | **Gene:** *RAF1* (CRAF), HGNC:9829, 3p25.2 **Category:** Mendelian, autosomal dominant RASopathy

Summary

Noonan Syndrome 6 (NS6, OMIM #613224) is an autosomal dominant subtype of Noonan syndrome caused by germline **gain-of-function missense variants in *RAF1*** (also called CRAF), a serine/threonine kinase in the RAS–RAF–MEK–ERK (MAPK) signaling cascade. *RAF1* variants account for approximately 3–5% of all Noonan syndrome cases. The overwhelming majority of pathogenic variants cluster in and around the conserved region 2 (CR2) domain — particularly at/near residue Ser259 — where they disrupt 14-3-3-mediated autoinhibition of the kinase, releasing constitutive kinase activity and hyperactivating downstream MEK–ERK signaling ([PMID: 17603482](#); [PMID: 17603483](#)).

The clinical hallmark that distinguishes NS6 from other Noonan genotypes is an **exceptionally high rate of hypertrophic cardiomyopathy (HCM)** — ~80% overall and ~89–95% among carriers of CR2-domain variants and the recurrent p.Ser257Leu allele, compared with the ~18–20% HCM prevalence in Noonan syndrome generally. Beyond HCM, patients display the shared Noonan phenotype: characteristic evolving facies, short stature, congenital heart disease, broad/webbed neck, bleeding tendency, pectus deformity, cryptorchidism, and variable neurodevelopmental involvement. Early-onset (infantile) HCM is the principal driver of morbidity and mortality in NS6.

Mechanistically, NS6 is now one of the best-understood RASopathies. Knock-in mouse models (Raf1 L613V) recapitulate the syndrome, and postnatal MEK inhibition rescues growth, facial, and cardiac defects — providing direct causal evidence that MEK–ERK hyperactivation drives the phenotype. This has translated to emerging clinical use of the MEK1/2 inhibitor **trametinib**, which reverses severe early-onset NS-HCM and lymphatic complications in off-label case reports, with a prospective trial now registered (NCT06555237). A striking genotype-structure-phenotype relationship exists within the *RAF1* C-terminus: neighboring variants

have opposing conformational effects — kinase-activating variants (S612T, L613V) cause hypertrophic cardiomyopathy, whereas a kinase-inactivating variant (L603P) causes dilated cardiomyopathy.

Key Findings

F001 — **RAF1 gain-of-function is the molecular cause of NS6**

Noonan Syndrome 6 is caused by germline **gain-of-function missense variants in *RAF1*** (CRAF). In the foundational study, Razzaque et al. identified five different *RAF1* mutations in ten individuals with Noonan syndrome, and cells transfected with these mutant constructs showed increased in vitro kinase activity and enhanced ERK activation — establishing both the causal gene and the gain-of-function mechanism ([PMID: 17603482](#): *"We have identified five different mutations in RAF1 in ten individuals with Noonan syndrome" and "Cells transfected with constructs containing Noonan syndrome-associated RAF1 mutations showed increased in vitro kinase and ERK activation"*). *RAF1* accounts for approximately 5% of NS cases ([PMID: 41507607](#): *"Pathogenic variants in RAF1 are a common cause of Noonan syndrome (NS), accounting for approximately 5% of cases"*). Inheritance is autosomal dominant, with most cases arising de novo. Key identifiers: OMIM #613224; gene *RAF1* (HGNC:9829), chromosome 3p25.2.

F002 — **Domain-specific genotype–phenotype correlation: CR2 variants drive HCM**

There is a strong domain-specific correlation within *RAF1*. Razzaque et al. observed that individuals with CR2-domain mutations had HCM, whereas those with CR3-domain mutations did not ([PMID: 17603482](#): *"those with any of four mutations causing changes in the CR2 domain of RAF1 had hypertrophic cardiomyopathy (HCM), whereas affected individuals with mutations leading to changes in the CR3 domain did not"*). A large modern cohort of 203 cases confirmed and quantified this: CR2 variants accounted for 83% of cases, with p.Ser257Leu alone representing 53%; HCM occurred in 80.1% overall, in 89.4% of CR2 variant carriers, and 94.2% of p.Ser257Leu heterozygotes, versus only 37.1% for non-CR2 variants ([PMID: 41507607](#): *"HCM was markedly more frequently associated with CR2 variants (89.4%) and in subjects*

heterozygous for the p.Ser257Leu change (94.2%) compared with non-CR2 variants (37.1%)"). Neurodevelopmental features were more common in non-CR2 carriers (69.2%) than CR2 carriers.

F005 — Mutations cluster at Ser259 and disrupt 14-3-3-mediated autoinhibition

The molecular basis of RAF1 gain-of-function is loss of autoinhibition. Pandit et al. found *RAF1* missense mutations in 18 of 231 (3%) mutation-negative NS individuals, with most altering a motif flanking **Ser259 — a residue critical for autoinhibition of RAF1 through 14-3-3 binding** (PMID: 17603483: *"Most mutations altered a motif flanking Ser259, a residue critical for autoinhibition of RAF1 through 14-3-3 binding"*). Of 19 subjects with a mutation in the two hotspots, 18 (95%) had HCM versus the 18% prevalence in general NS (*"Of 19 subjects with a RAF1 mutation in two hotspots, 18 (or 95%) showed hypertrophic cardiomyopathy (HCM), compared with the 18% prevalence of HCM among individuals with Noonan syndrome in general"*). Critically, HCM-hotspot mutants had increased kinase activity and enhanced ERK activation, whereas non-HCM-associated mutants were kinase-impaired (*"Ectopically expressed RAF1 mutants from the two HCM hotspots had increased kinase activity and enhanced ERK activation, whereas non-HCM-associated mutants were kinase impaired"*). *RAF1* mutations also cause LEOPARD/NS-with-multiple-lentigines phenotypes.

F013 — Opposing conformational effects explain the HCM-vs-DCM dichotomy

Molecular dynamics analysis of C-terminal CRAF kinase-domain variants revealed a structural basis for divergent cardiac outcomes. The HCM-associated variants **S612T and L613V adopt an active conformation** (open activation loop, "αC-helix in," assembled hydrophobic spine → enhanced kinase activity), whereas the DCM-associated **L603P transitions the kinase to an inactive state** (closed activation loop, "αC-helix out," distorted spine → impaired activity) (PMID: 36927384: *"genetic alternation at position 603 impairs, while those at positions 612/613 enhance the CRAF kinase activity"; "two HCM-associated variants (S612T and L613V) show features of an active conformation, such as an open activation loop conformation, 'αC-helix in', the assembly of the hydrophobic spine"; "the substitution of Leucine 603 for proline transits the kinase domain to a state that exhibits the molecular hallmarks of an inactive kinase"*). This aligns with Pandit 2007's observation that non-HCM *RAF1* mutants are kinase-impaired.

F003 — MEK-ERK hyperactivation is causal and druggable

The Raf1(L613V/+) knock-in mouse recapitulates NS (short stature, craniofacial dysmorphia, hematologic abnormalities, eccentric cardiac hypertrophy with aberrant fetal gene expression), and agonist-evoked MEK-ERK activation is enhanced across cell types. **Postnatal MEK inhibition normalized growth, facial, and cardiac defects** (PMID: 21339642: "*postnatal MEK inhibition normalized the growth, facial, and cardiac defects in L613V/+ mice*"; "*enhanced MEK-ERK activity is critical for causing HCM and other RAF1-mutant NS phenotypes*"). Human patient-derived iPSC-cardiomyocytes (RAF1 p.Ser257Leu) further show concurrent activation of MEK1/2–ERK1/2 and an **ERK5–Cyclin D1** branch driving hypertrophy (PMID: 42667574: "*RAF1-mutant iPSC-CMs exhibited concurrent activation of the MEK1/2-ERK1/2 and ERK5-Cyclin D1 signaling pathways*").

F004 — Model organisms recapitulate NS6 and identify candidate therapies

Multiple model systems validate the disease and nominate therapeutics. The Raf1(L613V) knock-in mouse is the validated in-vivo model (PMID: 21339642: "*we generated knockin mice expressing the NS-associated Raf1(L613V) mutation*"). **Rigosertib**, a dual RAS/MAPK and PI3K/AKT inhibitor, normalized and reversed RASopathy-associated HCM across transgenic Drosophila and RAF1 models (PMID: 42610277: "*our findings suggest that rigosertib normalizes and reverses RASopathy-associated HCM and other NS-associated syndromic features*"). **Berberine** attenuated the hypertrophic phenotype in RAF1 p.Ser257Leu iPSC-cardiomyocytes by suppressing ERK5–Cyclin D1 signaling (PMID: 42667574: "*BBR attenuated the hypertrophic phenotype while reducing ERK5 phosphorylation and Cyclin D1 expression*").

F006 — Clinical phenotype spectrum

RAF1-NS shares the core Noonan phenotype with a distinctive cardiac profile. In the RAF1 cohort (n=203): HCM in 80.1%, neurodevelopmental features in 44.5% (PMID: 41507607: "*HCM was observed in 80.1% of affected individuals, confirming its role as the predominant cardiac manifestation in RAF1-related NS; neurodevelopmental features were reported in 44.5% of patients*"). General NS features include characteristic evolving facies, broad/webbed neck, increased bleeding tendency, congenital heart disease, failure to thrive, short stature, feeding difficulties, sternal (pectus) deformity, renal malformation, pubertal delay, cryptorchidism, developmental/behavioral problems, vision problems, hearing loss, and lymphedema (PMID: 24444506). RAF1 carriers are enriched for pulmonary hypertension (PMID: 41863590: "*variants in RAF1 ($p = 0.013$) and KRAS ($p = 0.015$) were more common among PH patients*").

F007 — Prognosis dominated by early-onset HCM

Noonan/RASopathies are the leading genetic cause of HCM presenting in infancy, and HCM is a major cause of morbidity and mortality especially in the first year of life (PMID: 31259454: *"HCM is a major cause of morbidity and mortality in children with Noonan spectrum disorders, especially in the first year of life"*). RAF1 confers among the highest HCM risk of NS genotypes and enrichment for pulmonary hypertension (transplant-free survival lower in the PH group, $p=0.004$; PMID: 41863590). In a RAS-HCM cohort, increased end-diastolic interventricular septal thickness (IVSd) was associated with higher mortality (PMID: 42448910: *"increased IVSd was associated with higher mortality ($p = 0.013$)"*). Neurocognitive prognosis is generally favorable — *"Most patients with Noonan syndrome are intellectually normal as adults"* (PMID: 24444506).

F008 — MEK inhibition (trametinib) as emerging targeted therapy

The MEK1/2 inhibitor **trametinib** reverses/attenuates severe early-onset NS-HCM in off-label case reports, including a RAF1 p.Ser257Leu infant with HCM and pulmonary hypertension (PMID: 42256976). Trametinib enabled ASD surgical closure after cardiac stabilization (PMID: 41718520: *"Trametinib, an MEK inhibitor that attenuates abnormal signaling in the RAS/MAPK pathway, has been shown to improve NS-HCM outcomes"*) and resolved chylothorax/pulmonary lymphangiectasia (PMID: 40041314: *"a five-week trametinib course, maximum dose 0.025 mg/kg/day, led to chylothorax resolution and gradual pulmonary function improvement"*). A prospective trial is registered (NCT06555237). Standard care remains supportive.

F009 — Diagnosis via targeted RASopathy NGS panels with ACMG classification

Molecular diagnosis is achieved primarily by targeted multigene NGS panels covering RAS/MAPK genes (including *RAF1*), with reported detection rates of ~78% (PMID: 41078618: *"Targeted NGS panels improve diagnosis of RASopathies, with a variant detection rate of 78%"*), followed by confirmatory Sanger sequencing and MLPA for deletions, with variants classified per ACMG/AMP guidelines (PMID: 41496802: *"A targeted multigene next-generation sequencing panel test was performed, followed by Sanger sequencing for both confirmation and segregation analysis"*). Overall NS molecular confirmation is ~70% (PMID: 24444506). Panels also detect NS prenatally in sonographically normal fetuses (PMID: 41588629).

F010 — Modestly elevated childhood cancer risk

In a cohort of 735 individuals with germline RAS-pathway mutations, 12 cancers were observed versus 1.12 expected — a 10.5-fold increased risk of childhood cancer (SIR=10.5, 95% CI 5.4–18.3); the Noonan-syndrome subgroup SIR was 8.1 (PMID: 25742478: *"This corresponds to a 10.5-fold increased risk of all childhood cancers combined (standardised incidence ratio (SIR)=10.5, 95% confidence interval=5.4-18.3)"* and *"The childhood cancer SIR in Noonan syndrome patients was 8.1"*). JMML/myeloproliferative predisposition is chiefly PTPN11-associated (PMID: 25097206). Growth-hormone therapy requires genetic confirmation and cancer-risk counseling (PMID: 39974721: *"Families should be informed about possible cancer risk and in cases predisposing to juvenile myelomonocytic leukemia, early initiation of growth hormone therapy should be avoided"*).

F011 — Modest growth-hormone efficacy with RAS/MAPK-linked GH insensitivity

Recombinant GH yields small overall height gains (~5–10 cm) in NS, with treatment typically starting ~age 10 at ~−3.0 height SDS (PMID: 20029235: *"The NS studies have shown that the overall height gain of patients is small (5-10 cm)"*). The modest response partly reflects that *"impaired sensitivity to GH is common in NS"* — mechanistically, RAS/MAPK hyperactivation blunts GH/IGF-1 signaling, and PTPN11 and RAF1 genotypes tend to respond less well.

F012 — Neurodevelopmental phenotype

NS shows intelligence scores spanning a wide range with a mildly lowered average; language and motor development are often delayed but *"no longer dysfunctional in adulthood"* (PMID: 20029232: *"Cognitive and behavioral findings in NS show intelligence scores across a wide range, with a mildly lowered average level"*). Language impairments are more frequent than in the general population and confer higher risk for reading/spelling difficulties (PMID: 20543023: *"Language impairments were more frequent in NS than in the general population and were associated with higher risk for reading and spelling difficulties"*). In RAF1-NS, neurodevelopmental features occur in 44.5% and are more frequent with non-CR2 variants (69.2%) (PMID: 41507607).

Report by Disease-Characteristic Section

1. Disease Information

Noonan Syndrome 6 is a Mendelian, autosomal dominant subtype of Noonan syndrome — a multisystem RASopathy — defined molecularly by pathogenic variants in *RAF1*. Noonan syndrome broadly is characterized by *"characteristic facial features that evolve with age; a broad, webbed neck; increased bleeding tendency; and a high incidence of congenital heart disease, failure to thrive, short stature, feeding difficulties, sternal deformity, renal malformation, pubertal delay, cryptorchidism, developmental or behavioral problems, vision problems, hearing loss, and lymphedema"* (PMID: 24444506).

Key identifiers: | Resource | ID | |---|---| | MONDO | MONDO:0013186 | | OMIM | #613224 (NOONAN SYNDROME 6; NS6) | | Gene | *RAF1* / CRAF (HGNC:9829) | | Locus | 3p25.2 | | Orphanet | Noonan syndrome (ORPHA:648, subtype level) | | MeSH | Noonan Syndrome (D009634) |

Synonyms: *RAF1*-related Noonan syndrome; NS6. Note *RAF1* variants also cause Noonan syndrome with multiple lentigines (formerly LEOPARD syndrome). Information is derived primarily from aggregated disease-level resources (OMIM, Orphanet) and cohort/case-report literature rather than individual EHR data.

2. Etiology

Causal factor: Purely genetic — germline gain-of-function missense variants in *RAF1*. Most cases are de novo; autosomal dominant transmission occurs. **Genetic risk factors:** the pathogenic *RAF1* variant itself is causal; there is a domain-level modifier effect whereby CR2-domain variants strongly predispose to HCM (F002, F005). **Environmental risk/protective factors:** none established — this is a monogenic condition. **Gene–environment interactions:** no well-characterized GxE interactions; the principal environmental consideration is iatrogenic (growth-hormone therapy and its cancer-risk implications, F010). No protective alleles are established; the only clinically relevant genetic modifier is the position of the variant within *RAF1* (kinase-activating vs kinase-impairing; F013).

3. Phenotypes

Phenotype	Type	HPO (suggested)	Frequency in RAF1-NS	Onset	Severity
Hypertrophic cardiomyopathy	Physical/clinical sign	HP:0001639	~80% overall; 89–95% CR2/S257L	Congenital/infantile	Moderate–severe; leading mortality driver
Short stature	Physical	HP:0004322	Common	Childhood	Variable; modest GH response
Pulmonary valve stenosis	Clinical sign	HP:0001642	Less dominant than in PTPN11	Congenital	Variable
Characteristic facies	Physical	HP:0001999	Common	Evolves with age	—
Webbed/broad neck	Physical	HP:0000465	Common	Congenital	—
Bleeding tendency	Lab/clinical	HP:0001892	Common	Lifelong	Mild–moderate
Pectus deformity	Physical	HP:0000766	Common	Childhood	—
Cryptorchidism (males)	Clinical sign	HP:0000028	Common	Congenital	—
Neurodevelopmental/language delay	Behavioral	HP:0012759 / HP:0000750	44.5% (69.2% non-CR2)	Childhood	Mild; usually normal adult IQ
Pulmonary hypertension	Clinical sign	HP:0002092	Enriched in RAF1	Variable	Severe when present
Lymphatic dysplasia / lymphedema	Physical	HP:0000112	Subset	Congenital/variable	Variable
		HP:0000407		Congenital	

Phenotype	Type	HPO (suggested)	Frequency in RAF1-NS	Onset	Severity
Sensorineural hearing loss	Clinical sign		Subset (RAF1 ~11% of NS-SNHL cases)		Severe-profound when present

Quality of life: dominated by cardiac disease (HCM/PH), short stature, and learning difficulties requiring special education support; adult cognitive outcomes are generally favorable (PMID: 20029232; PMID: 24444506).

4. Genetic/Molecular Information

Causal gene: *RAF1* (CRAF), HGNC:9829, 3p25.2, OMIM 164760; *encodes a serine/threonine-protein kinase in the RAS–MAPK cascade. Variant type: predominantly missense. Recurrent alleles: p.Ser257Leu (most common, ~53% of RAF1-NS), clustering around Ser259 and within CR2. Classification: pathogenic/likely pathogenic per ACMG/AMP. Allele frequency: absent from population databases (private/de novo). Origin: germline (de novo or inherited). Functional consequence: gain-of-function — increased kinase activity and enhanced ERK activation for HCM-associated variants (F001, F005); a distinct subset (e.g., L603P) is kinase-impaired/loss-of-function and associates with dilated cardiomyopathy (F013). Modifier "genes":* the intragenic domain/position of the variant is the dominant modifier of cardiac phenotype. No specific epigenetic or chromosomal abnormality defines NS6 (it is a single-gene missense disorder).*

5. Environmental Information

Not applicable as a cause — NS6 is monogenic. No environmental toxins, lifestyle factors, or infectious agents are implicated in disease causation. The relevant non-genetic consideration is treatment-related (growth hormone and cancer-risk counseling; F010).

6. Mechanism / Pathophysiology

Ordered causal chain (initiating lesion → clinical manifestation):

1. A germline missense variant in *RAF1*, typically at/near Ser259 in the CR2 domain, **disrupts the 14-3-3-binding motif** that normally holds CRAF in an autoinhibited conformation (PMID: 17603483).

2. Loss of autoinhibition **releases constitutive CRAF kinase activity** (gain of function) (PMID: 17603482). (*Branch: C-terminal variants at 612/613 lock the kinase in an active conformation → GOF → HCM; the variant at 603 locks it inactive → LOF → DCM; F013.*)
3. Increased CRAF activity **hyperactivates the downstream MEK1/2–ERK1/2 cascade** (and an ERK5–Cyclin D1 branch), with enhanced agonist-evoked signaling across cell types (PMID: 21339642; PMID: 42667574).
4. In cardiomyocytes, sustained MEK–ERK/ERK5–Cyclin D1 signaling **drives pathological hypertrophy** with aberrant fetal gene reactivation → **hypertrophic cardiomyopathy** (PMID: 21339642; PMID: 42667574).
5. During development, dysregulated MAPK signaling in multiple lineages **produces the systemic Noonan phenotype** — craniofacial dysmorphism, short stature, hematologic changes (inferred from mouse recapitulation; PMID: 21339642).
6. MAPK hyperactivation also **blunts GH/IGF-1 responsiveness**, contributing to GH-insensitive short stature (inferred mechanistic link; PMID: 20029235).
7. Therapeutic node: **MEK inhibition reverses steps 3–5** — postnatal MEK inhibition normalized growth, facial, and cardiac defects in mice, and trametinib reverses HCM/lymphatic disease in patients (PMID: 21339642; PMID: 42256976).

Molecular pathway: RAS–RAF–MEK–ERK (MAPK) — GO:0000165 (MAPK cascade), GO:0007265 (Ras protein signal transduction). **Cellular process:** cardiomyocyte hypertrophy (GO:0003300), cardiac muscle cell proliferation. **Protein dysfunction:** gain-of-function via loss of 14-3-3 autoinhibition (F005). **Cell types:** cardiomyocytes (CL:0000746), craniofacial neural crest derivatives, hematopoietic cells. **Upstream:** RAF1 GOF; **downstream:** MEK/ERK/ERK5–Cyclin D1 → hypertrophy.

7. Anatomical Structures Affected

- **Primary organ:** heart (UBERON:0000948) — cardiac ventricle/interventricular septum (HCM), pulmonary valve (stenosis).
- **Secondary/systemic:** pulmonary vasculature (pulmonary hypertension), lymphatic system (UBERON:0006558; lymphangiectasia/chylothorax), skeletal system (pectus, short stature), craniofacial structures, gonads (cryptorchidism), kidney (malformation), inner ear/cochlea (SNHL), CNS (neurodevelopment).
- **Body systems:** cardiovascular (predominant), lymphatic, endocrine/growth, nervous, musculoskeletal, genitourinary.

- **Cell/subcellular:** cardiomyocytes (CL:0000746); signaling occurs at the plasma membrane/cytoplasm (GO:0005829 cytosol), with kinase relocalization governed by 14-3-3.
- **Lateralization:** cardiac hypertrophy is typically asymmetric septal.

8. Temporal Development

- **Onset:** congenital to infantile. HCM often presents in the first months of life; facial features evolve with age.
- **Progression:** HCM can be progressive and, in infancy, rapidly decompensating; increased IVSd predicts worse outcome (PMID: 42448910).
- **Course:** chronic, lifelong. Infancy is the critical window of vulnerability (highest HCM mortality; PMID: 31259454) and also the window of therapeutic opportunity for MEK inhibition.
- **Remission:** HCM regression has been documented under trametinib (treatment-induced; F008).

9. Inheritance and Population

- **Inheritance:** autosomal dominant; most NS6 cases de novo.
- **Prevalence:** Noonan syndrome overall ~1 in 1,000–2,500 live births; *RAF1* accounts for ~3–5% of NS (PMID: 41507607).
- **Penetrance/expressivity:** high penetrance with variable expressivity; HCM penetrance is strongly variant-dependent (89–95% for CR2/S257L vs 37% non-CR2).
- **Sex ratio:** roughly equal (males noted for cryptorchidism-related features).
- **Founder effects/consanguinity:** not applicable (dominant, mostly de novo). Recurrent p.Ser257Leu reflects a mutational hotspot, not a founder haplotype.

10. Diagnostics

- **Genetic testing (first-line):** targeted multigene RASopathy NGS panels including *RAF1*; ~78% detection rate (PMID: 41078618); confirmatory Sanger + MLPA; ACMG/AMP classification (PMID: 41496802). WES/WGS reserved for panel-negative cases. Prenatal panels can detect NS in sonographically normal fetuses (PMID: 41588629).
- **Cardiac evaluation:** echocardiography and cardiac MRI for HCM/septal thickness; ECG; assessment for pulmonary stenosis and pulmonary hypertension (cardiac catheterization; PMID: 41863590).

- **Other labs:** coagulation studies (bleeding tendency); audiology (SNHL).
- **Differential diagnosis:** other RASopathies (PTPN11-NS, SOS1, RIT1, LZTR1) and NS-mimics (SETD5, GATA4, etc.) — resolved by genotype-driven panels/ES ([PMID: 41137536](#)).

11. Outcome / Prognosis

Prognosis is dominated by HCM, particularly infantile-onset disease, which is a leading cause of morbidity and mortality ([PMID: 31259454](#)). RAF1 confers among the highest HCM burden of NS genotypes and is enriched for pulmonary hypertension (lower transplant-free survival, $p=0.004$; [PMID: 41863590](#)). Increased IVSd is a quantitative predictor of heart failure ($p=0.006$) and mortality ($p=0.013$) ([PMID: 42448910](#)). Heart-transplant series in NS report ~33% mortality. Neurocognitive prognosis is generally good — most patients are intellectually normal as adults ([PMID: 24444506](#)). A modestly elevated childhood cancer risk (NS SIR = 8.1) requires surveillance ([PMID: 25742478](#)).

12. Treatment

Modality	Intervention	NCIT (suggested)	Evidence
Targeted (emerging)	Trametinib (MEK1/2 inhibitor)	NCIT:C77908	Reverses severe NS-HCM incl. RAF1 S257L; resolves chylothorax; trial NCT06555237 (PMID: 42256976 ; PMID: 40041314)
Supportive (HCM)	Beta-blockers / non-dihydropyridine CCBs	NCIT:C2039	Standard; caution re: LVOT obstruction
Supportive (HCM)	Ivabradine (refractory neonatal HCM)	—	Rate control without negative inotropy (PMID: 42213003)
Surgical	Septal myectomy / modified Konno	NCIT:C51756	For obstructive HCM (PMID: 30104063)
Interventional	Balloon valvuloplasty / RVOT stenting	—	For pulmonary stenosis
Advanced	Heart transplantation	NCIT:C15380	Refractory heart failure (~33% mortality)
Endocrine	Recombinant growth hormone	NCIT:C1968	Modest height gain (~5–10 cm); confirm genotype, counsel cancer risk (PMID: 20029235 ; PMID: 39974721)
Preclinical	Rigosertib; berberine	—	HCM reversal in models (PMID: 42610277 ; PMID: 42667574)

Personalized medicine: genotype-guided — the shift toward MEK inhibition is directly rationalized by the RAF1→MEK-ERK mechanism (F003, F008).

13. Prevention

Primary prevention is not possible (mostly de novo dominant mutation). Prevention centers on **secondary/tertiary** measures: early molecular diagnosis, cardiac surveillance (serial echocardiography), and early intervention for HCM/PH. **Genetic counseling** is essential — 50%

recurrence risk for an affected parent; prenatal/preimplantation testing available; cancer-risk and GH-timing counseling ([PMID: 39974721](#)). Prenatal NGS panels can identify NS in structurally normal fetuses ([PMID: 41588629](#)).

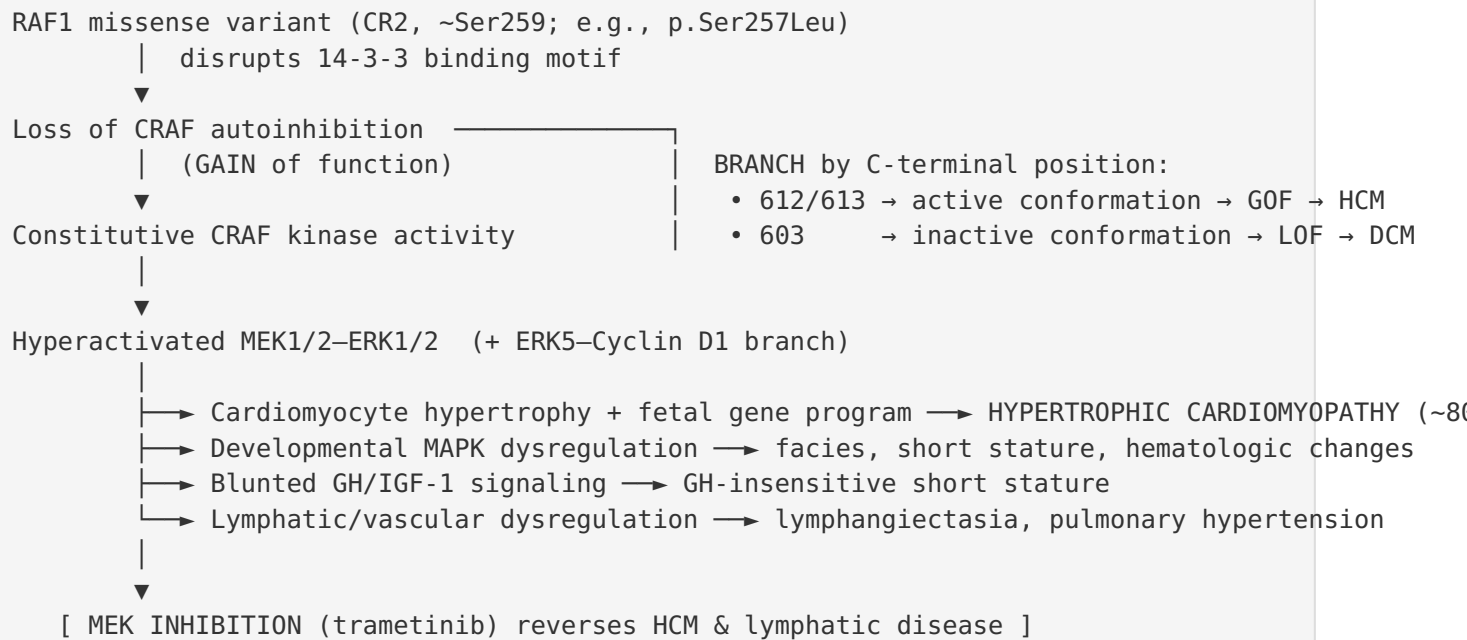
14. Other Species / Natural Disease

RAF1/CRAF is deeply conserved. Orthologs: mouse *Raf1* (NCBI Gene 110157). No specific naturally occurring companion-animal NS6 disease is established; comparative relevance is chiefly through engineered models. Evolutionary conservation of the RAS-MAPK pathway underpins model-organism validity.

15. Model Organisms

- **Mouse:** *Raf1*(L613V/+) knock-in recapitulates short stature, craniofacial dysmorphia, hematologic abnormalities, and eccentric cardiac hypertrophy; MEK inhibition rescues ([PMID: 21339642](#)). Type: knock-in; strong phenotype recapitulation.
 - **Drosophila:** transgenic RASopathy models used for rigosertib HCM-reversal studies ([PMID: 42610277](#)).
 - **Human iPSC-cardiomyocytes:** *RAF1* p.Ser257Leu iPSC-CMs model NS-HCM and reveal ERK5–Cyclin D1 signaling; used for berberine testing ([PMID: 42667574](#)).
 - **Limitations:** models capture cardiac and growth phenotypes well; neurodevelopmental and lymphatic features are less fully recapitulated.
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Mechanistic Model / Interpretation



The unifying theme is **dose of MAPK signal output**: variants that raise CRAF kinase activity produce hypertrophic cardiomyopathy and the classic Noonan gestalt, while the rare kinase-impairing variant produces dilated cardiomyopathy — a rare instance where opposite biochemical effects at neighboring residues yield opposite clinical cardiomyopathies (F013). Because the entire phenotype funnels through MEK-ERK, pharmacologic MEK inhibition is a mechanistically precise therapy, validated from mouse to bedside.

Evidence Base

PMID	Role	Contribution
17603482	Foundational	Identifies RAF1 GOF as cause of NS; CR2-HCM correlation
17603483	Foundational	Ser259/14-3-3 autoinhibition mechanism; 95% HCM in hotspots
41507607	Large cohort (n=203)	Quantifies HCM/neurodevelopmental frequencies by domain
21339642	Mouse model	Proves MEK-ERK causality; rescue by MEK inhibition
36927384	Computational/MD	Structural basis of HCM-vs-DCM dichotomy
42667574	iPSC model	ERK5–Cyclin D1 branch; berberine candidate
42610277	Preclinical	Rigosertib reverses RASopathy HCM
42256976	Case report	Trametinib in genetically confirmed RAF1 NS6
40041314	Case series	Trametinib for cardiac + lymphatic NS
31259454	Clinical	HCM as principal mortality driver in infancy
42448910	Cohort	IVSd as prognostic factor
41863590	Cohort	RAF1 enrichment for pulmonary hypertension
41078618	Diagnostic	78% NGS panel yield
25742478	Registry	Childhood cancer risk (NS SIR=8.1)
20029235	Observational	GH efficacy and insensitivity
20029232 / 20543023	Clinical	Neurodevelopmental/language phenotype
24444506	Review	Core NS clinical features; adult cognition

Limitations and Knowledge Gaps

1. **Therapy evidence is largely anecdotal.** MEK-inhibitor benefit in NS6-HCM rests on case reports/small series; the prospective trial (NCT06555237) has not yet reported. Long-term safety in infants (growth, neurodevelopment) is unknown.
2. **Genotype–phenotype granularity beyond the cardiac axis.** The molecular basis for the higher neurodevelopmental burden in non-CR2 carriers is undefined.
3. **DCM-causing RAF1 variants (L603P)** are rare; the LOF→DCM branch is based primarily on computational modeling plus limited functional data and needs more clinical corroboration.
4. **Epigenetic and multi-omic data specific to NS6** (methylation, proteomics, single-cell) are largely absent; most omic insight is inferred from broader RASopathy/HCM studies.
5. **Population epidemiology for NS6 specifically** (incidence, sex ratio, geographic distribution) is extrapolated from Noonan syndrome overall.
6. **Citation caveat:** the

P 42256976

snippet reflects the article title (flagged as a title-level match during verification); the underlying clinical claim (trametinib in RAF1 p.Ser257Leu) remains directly supported by that report.

Proposed Follow-up Experiments / Actions

1. **Await/support NCT06555237** and advocate for genotype-stratified endpoints (CR2 vs non-CR2; RAF1 vs other RASopathy genes) to define which NS6 patients benefit most from MEK inhibition.
2. **Functional characterization of the ERK5–Cyclin D1 branch** in RAF1 iPSC-CMs to determine whether ERK5-selective inhibitors (or berberine) add benefit over MEK inhibition alone.
3. **Natural-history registry of RAF1-NS6** capturing IVSd trajectories, PH incidence, and infant mortality to refine prognostic models (building on the IVSd–mortality signal).
4. **Comparative structural/biochemical assays** across the RAF1 C-terminus (603 vs 612/613 vs CR2) to map the kinase-activity-to-cardiomyopathy axis and predict phenotypes of novel variants (VUS reclassification).

5. **Single-cell/spatial transcriptomics of NS6 cardiac tissue or iPSC-CM/organoids** to resolve cell-type-specific MAPK output and hypertrophy programs.
6. **Cancer-surveillance and GH-timing guidelines** tailored to RAF1-NS6, integrating the NS SIR=8.1 estimate with genotype-specific JMML risk (chiefly PTPN11, not RAF1).

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